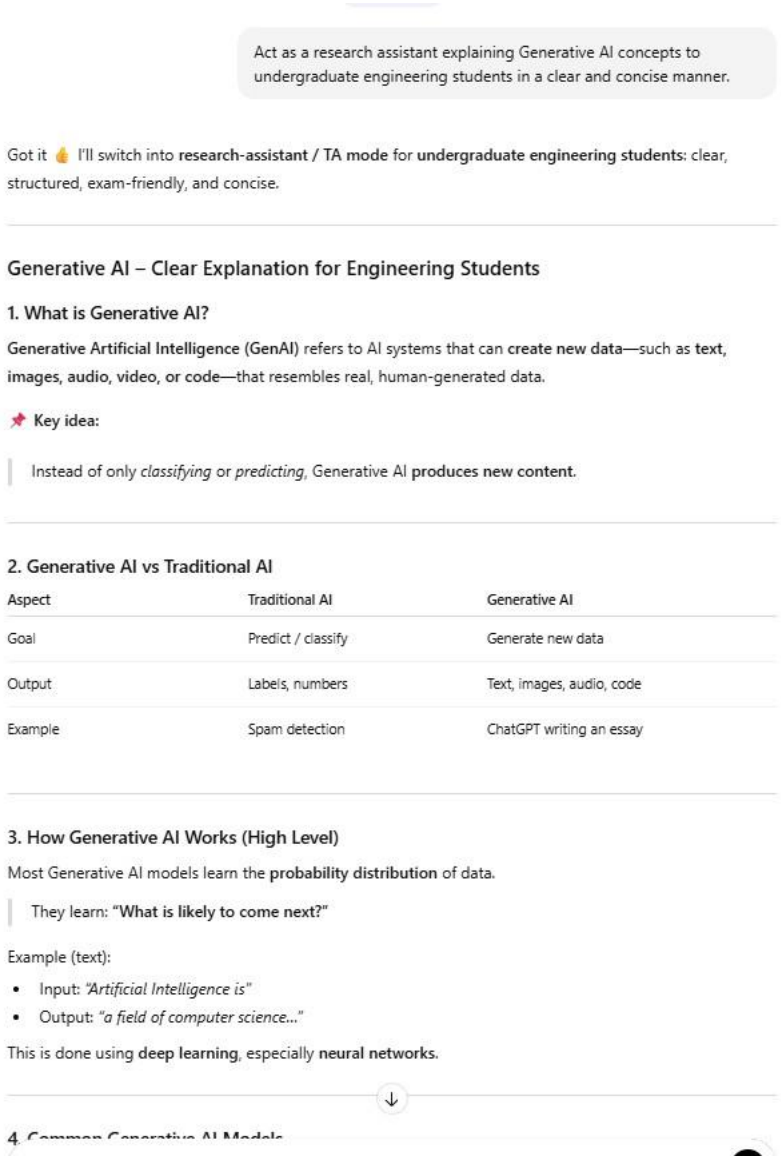
Lab Experiment 1 Activity

1. Initial Interactions: ChatGPT, Gemini, Perplexity, and Claude

- Understanding that while various LLMs share similarities, their outputs can vary based on their training and search capabilities.
- **ChatGpt**



Act as a research assistant explaining Generative AI concepts to undergraduate engineering students in a clear and concise manner.

Got it 👍 I'll switch into **research-assistant / TA mode** for **undergraduate engineering students**: clear, structured, exam-friendly, and concise.

Generative AI – Clear Explanation for Engineering Students

1. What is Generative AI?

Generative Artificial Intelligence (GenAI) refers to AI systems that can **create new data**—such as **text, images, audio, video, or code**—that resembles real, human-generated data.

📌 **Key idea:**

> Instead of only *classifying* or *predicting*, Generative AI **produces new content**.

---

2. Generative AI vs Traditional AI

Aspect	Traditional AI	Generative AI
Goal	Predict / classify	Generate new data
Output	Labels, numbers	Text, images, audio, code
Example	Spam detection	ChatGPT writing an essay

---

3. How Generative AI Works (High Level)

Most Generative AI models learn the **probability distribution** of data.

> They learn: **"What is likely to come next?"**

Example (text):

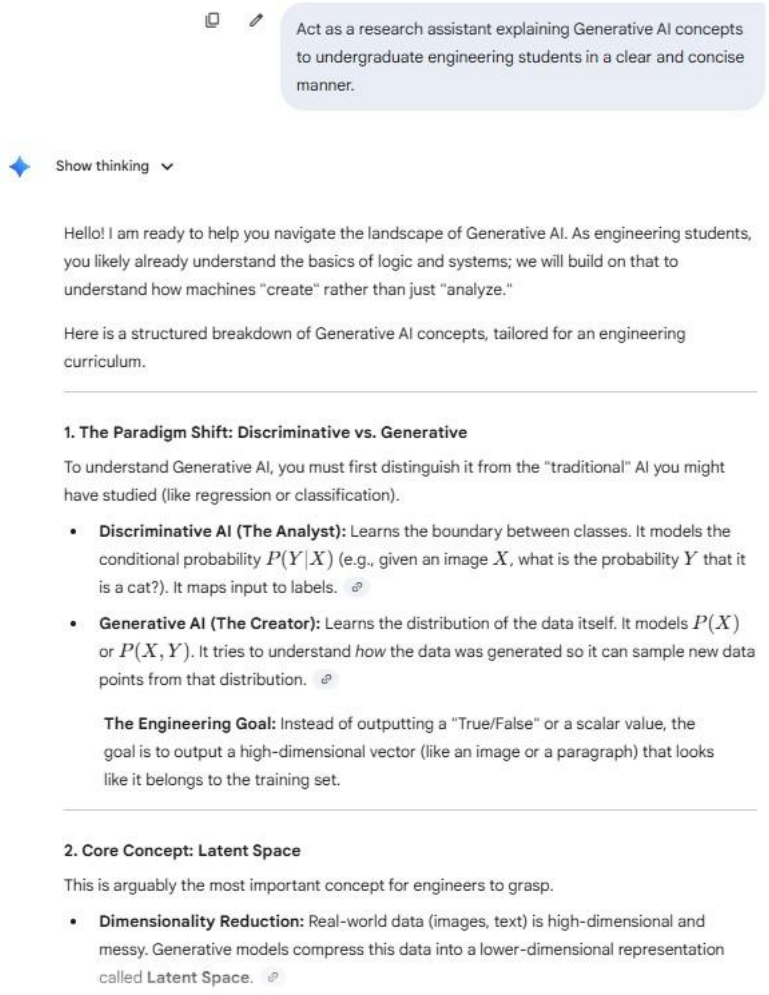
- Input: *"Artificial Intelligence is"*
- Output: *"a field of computer science..."*

This is done using **deep learning**, especially **neural networks**.

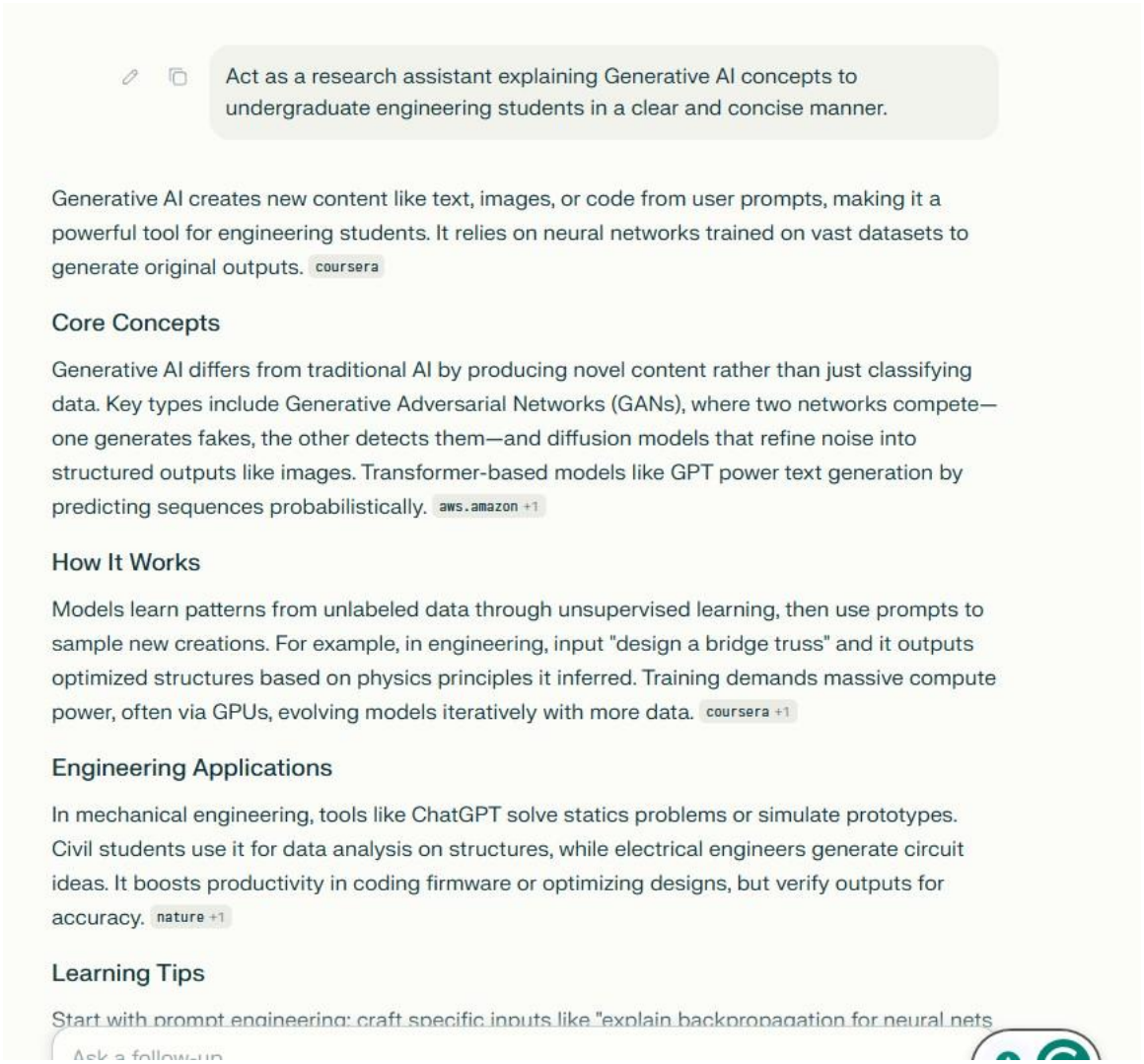
---

4. Common Generative AI Models

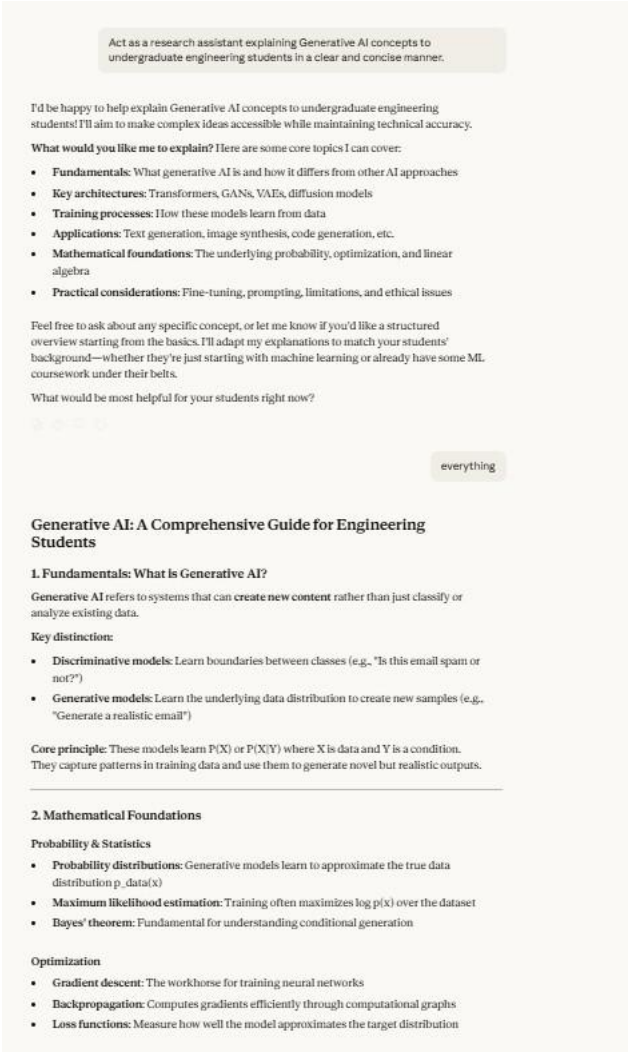
- **Gemini**


Act as a research assistant explaining Generative AI concepts to undergraduate engineering students in a clear and concise manner.
Show thinking
Hello! I am ready to help you navigate the landscape of Generative AI. As engineering students, you likely already understand the basics of logic and systems; we will build on that to understand how machines "create" rather than just "analyze."
Here is a structured breakdown of Generative AI concepts, tailored for an engineering curriculum.
1. The Paradigm Shift: Discriminative vs. Generative
To understand Generative AI, you must first distinguish it from the "traditional" AI you might have studied (like regression or classification).
Discriminative AI (The Analyst): Learns the boundary between classes. It models the conditional probability $P(Y|X)$ (e.g., given an image X, what is the probability Y that it is a cat?). It maps input to labels.
Generative AI (The Creator): Learns the distribution of the data itself. It models $P(X)$ or $P(X, Y)$. It tries to understand how the data was generated so it can sample new data points from that distribution.
The Engineering Goal: Instead of outputting a "True/False" or a scalar value, the goal is to output a high-dimensional vector (like an image or a paragraph) that looks like it belongs to the training set.
2. Core Concept: Latent Space
This is arguably the most important concept for engineers to grasp.
Dimensionality Reduction: Real-world data (images, text) is high-dimensional and messy. Generative models compress this data into a lower-dimensional representation called Latent Space.

- **Perplexity**


Act as a research assistant explaining Generative AI concepts to undergraduate engineering students in a clear and concise manner.
Generative AI creates new content like text, images, or code from user prompts, making it a powerful tool for engineering students. It relies on neural networks trained on vast datasets to generate original outputs. coursera
Core Concepts
Generative AI differs from traditional AI by producing novel content rather than just classifying data. Key types include Generative Adversarial Networks (GANs), where two networks compete—one generates fakes, the other detects them—and diffusion models that refine noise into structured outputs like images. Transformer-based models like GPT power text generation by predicting sequences probabilistically. aws.amazon +1
How It Works
Models learn patterns from unlabeled data through unsupervised learning, then use prompts to sample new creations. For example, in engineering, input "design a bridge truss" and it outputs optimized structures based on physics principles it inferred. Training demands massive compute power, often via GPUs, evolving models iteratively with more data. coursera +1
Engineering Applications
In mechanical engineering, tools like ChatGPT solve statics problems or simulate prototypes. Civil students use it for data analysis on structures, while electrical engineers generate circuit ideas. It boosts productivity in coding firmware or optimizing designs, but verify outputs for accuracy. nature +1
Learning Tips
Start with prompt engineering: craft specific inputs like "explain backpropagation for neural nets

- **Claude**



Act as a research assistant explaining Generative AI concepts to undergraduate engineering students in a clear and concise manner.

I'd be happy to help explain Generative AI concepts to undergraduate engineering students! I'll aim to make complex ideas accessible while maintaining technical accuracy.

What would you like me to explain? Here are some core topics I can cover:

- **Fundamentals:** What generative AI is and how it differs from other AI approaches
- **Key architectures:** Transformers, GANs, VAEs, diffusion models
- **Training processes:** How these models learn from data
- **Applications:** Text generation, image synthesis, code generation, etc.
- **Mathematical foundations:** The underlying probability, optimization, and linear algebra
- **Practical considerations:** Fine-tuning, prompting, limitations, and ethical issues

Feel free to ask about any specific concept, or let me know if you'd like a structured overview starting from the basics. I'll adapt my explanations to match your students' background—whether they're just starting with machine learning or already have some ML coursework under their belts.

What would be most helpful for your students right now?

everything

Generative AI: A Comprehensive Guide for Engineering Students

1. Fundamentals: What is Generative AI?

Generative AI refers to systems that can **create new content** rather than just classify or analyze existing data.

Key distinction:

- **Discriminative models:** Learn boundaries between classes (e.g., "Is this email spam or not?")
- **Generative models:** Learn the underlying data distribution to create new samples (e.g., "Generate a realistic email")

Core principle: These models learn P(X) or P(X|Y) where X is data and Y is a condition. They capture patterns in training data and use them to generate novel but realistic outputs.

---

2. Mathematical Foundations

Probability & Statistics

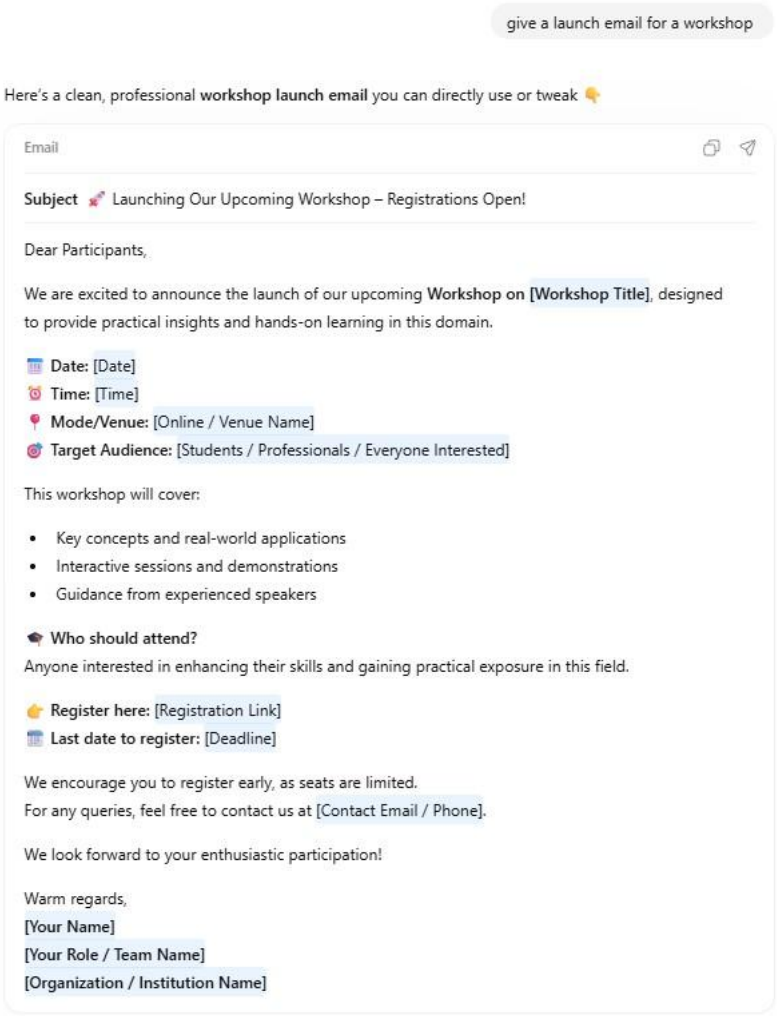
- **Probability distributions:** Generative models learn to approximate the true data distribution p_data(x)
- **Maximum likelihood estimation:** Training often maximizes log p(x) over the dataset
- **Bayes' theorem:** Fundamental for understanding conditional generation

Optimization

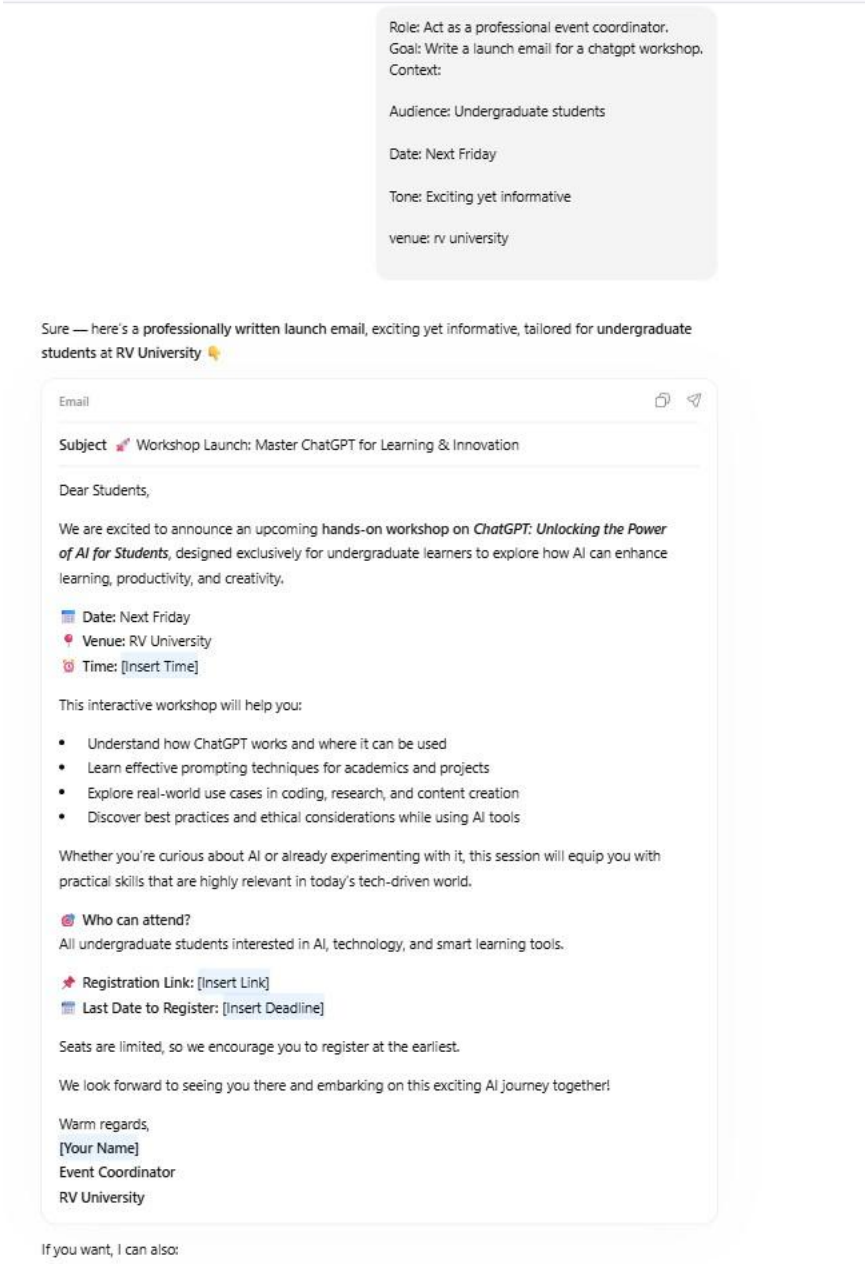
- **Gradient descent:** The workhorse for training neural networks
- **Backpropagation:** Computes gradients efficiently through computational graphs
- **Loss functions:** Measure how well the model approximates the target distribution

2. Practical Prompting: Launch Email for a Workshop

- Using structured frameworks to generate professional, ready-to-use content.



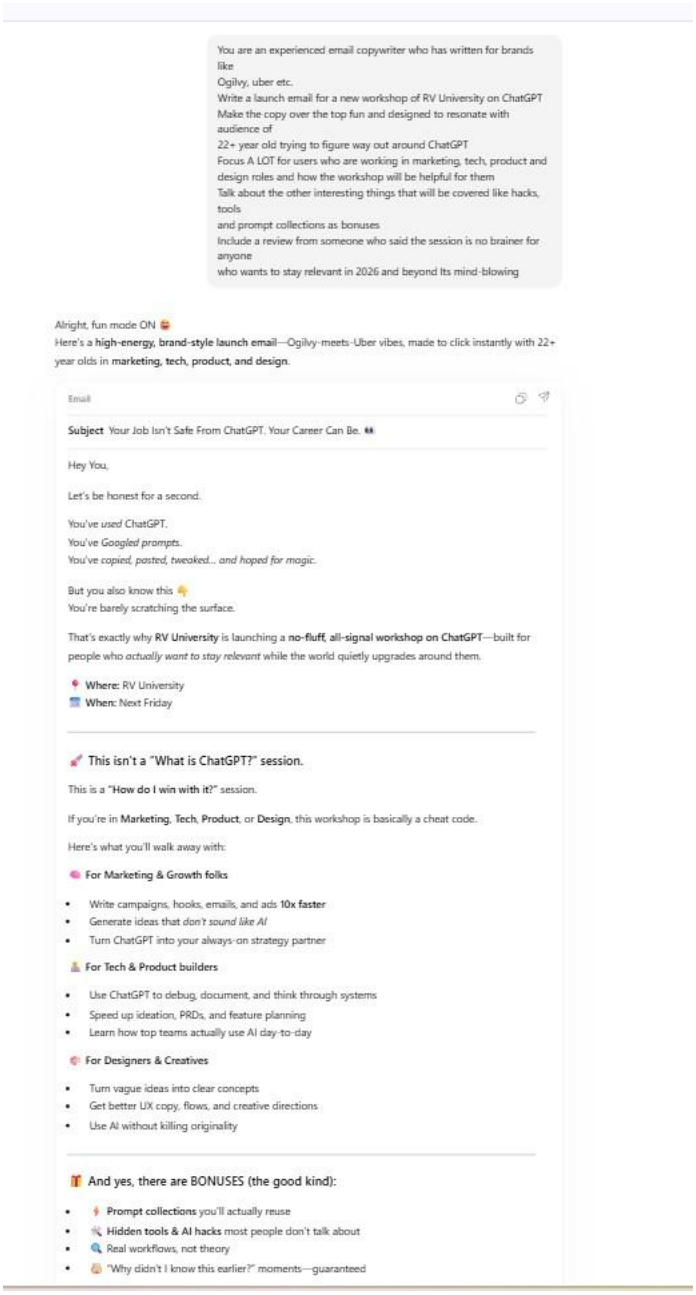
- Previous prompt was refined for a better content



Gagana.K
1RVU23CSE162

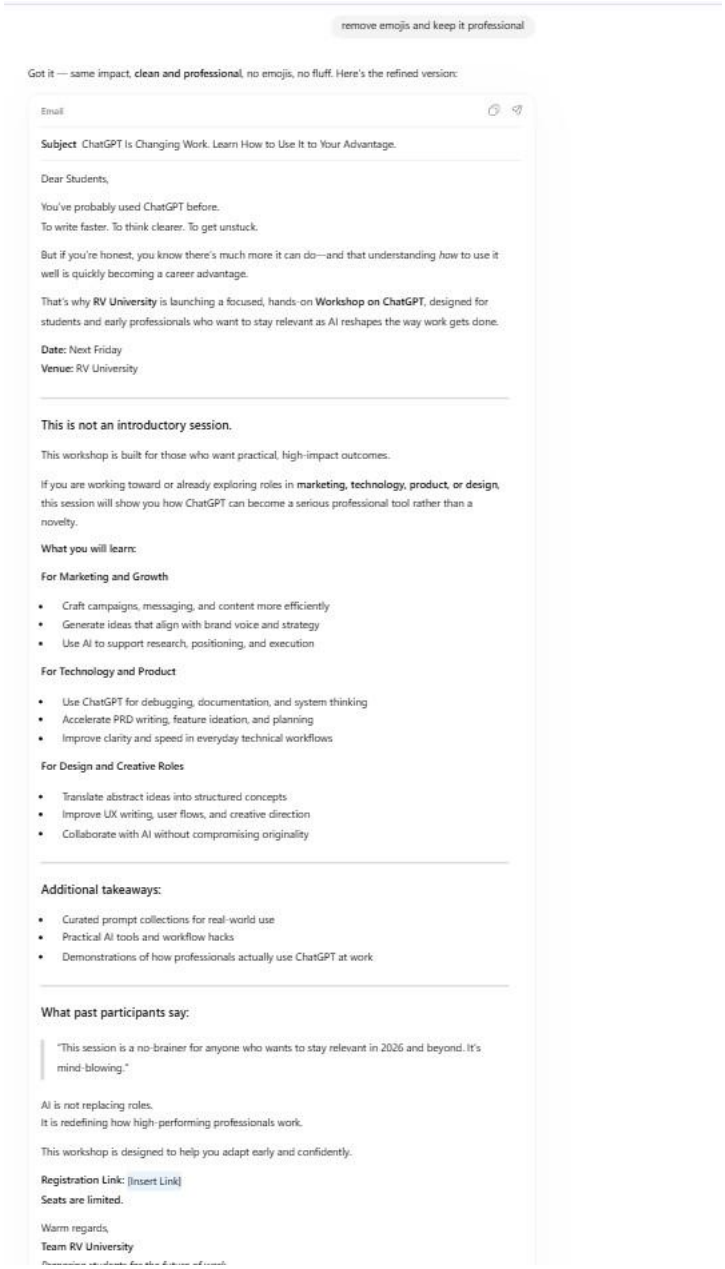
-

More information and context is given for a desired output



•

Refined to make the email professional and concise



remove emojis and keep it professional

Got it — same impact, **clean and professional**, no emojis, no fluff. Here's the refined version:

Email

Subject ChatGPT Is Changing Work. Learn How to Use It to Your Advantage.

Dear Students,

You've probably used ChatGPT before.
To write faster. To think clearer. To get unstuck.

But if you're honest, you know there's much more it can do—and that understanding *how* to use it well is quickly becoming a career advantage.

That's why **RV University** is launching a focused, hands-on **Workshop on ChatGPT**, designed for students and early professionals who want to stay relevant as AI reshapes the way work gets done.

Date: Next Friday
Venue: RV University

This is not an introductory session.

This workshop is built for those who want practical, high-impact outcomes.

If you are working toward or already exploring roles in **marketing, technology, product, or design**, this session will show you how ChatGPT can become a serious professional tool rather than a novelty.

What you will learn:

For Marketing and Growth

- Craft campaigns, messaging, and content more efficiently
- Generate ideas that align with brand voice and strategy
- Use AI to support research, positioning, and execution

For Technology and Product

- Use ChatGPT for debugging, documentation, and system thinking
- Accelerate PRD writing, feature ideation, and planning
- Improve clarity and speed in everyday technical workflows

For Design and Creative Roles

- Translate abstract ideas into structured concepts
- Improve UX writing, user flows, and creative direction
- Collaborate with AI without compromising originality

Additional takeaways:

- Curated prompt collections for real-world use
- Practical AI tools and workflow hacks
- Demonstrations of how professionals actually use ChatGPT at work

---

What past participants say:

> "This session is a no-brainer for anyone who wants to stay relevant in 2026 and beyond. It's mind-blowing."

AI is not replacing roles.
It is redefining how high-performing professionals work.

This workshop is designed to help you adapt early and confidently.

Registration Link: [Insert Link]
Seats are limited.

Warm regards,
Team RV University

Gagana.K
1RVU23CSE162

make email shorter , so it is informative and concise

Here's a **short, crisp, and professional** version—informative without losing impact:

Email

Subject Workshop Launch: Using ChatGPT Effectively for Your Career

Dear Students,

ChatGPT is rapidly changing how work gets done across marketing, technology, product, and design roles. Knowing *how* to use it effectively is quickly becoming a key professional skill.

RV University is pleased to launch a **hands-on workshop on ChatGPT**, focused on practical use cases that go beyond basic prompts.

Date: Next Friday
Venue: RV University

What you'll gain:

- Practical workflows for marketing, tech, product, and design roles
- Prompt strategies that improve speed, clarity, and quality of output
- Useful tools, hacks, and curated prompt collections you can reuse
- Real examples of how professionals use ChatGPT at work

What participants say:

> "This session is a no-brainer for anyone who wants to stay relevant in 2026 and beyond. It's mind-blowing."

Seats are limited.
Register here: [Insert Link]

Warm regards,
Team RV University

If you want it even tighter (or converted into a poster / WhatsApp version), I can do that too.

3. Image Generation:

• **Prompt:** generate an image where a beautiful woman is in rose garden, feeling the air wearing yellow dress.

[Kalaido.ai]



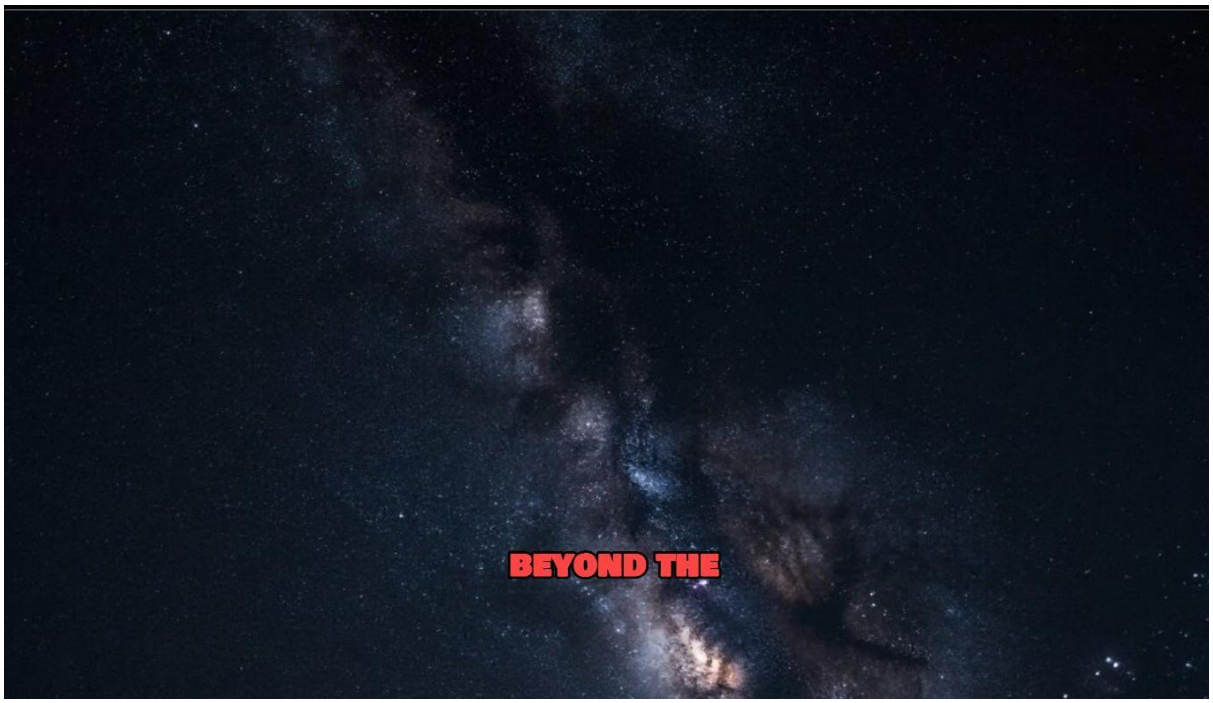


4. Video generation:

• Example for Chain of thoughts (for video generation):
- Astronomy Sci-Fi Video Script:
- Scene 1: Opening (Universe View) For thousands of years, humans looked at the night sky and wondered what lies beyond the stars.
- Scene 2: Solar System Exploration from the Moon to Mars, from Saturn's rings to Jupiter's storms, our curiosity pushed us deeper into space.
- Scene 3: Future Space Travel In the future, space stations orbit distant planets, and humans travel across galaxies.
- Scene 4: Cosmic Mystery Are we alone in the universe? Or is intelligent life waiting among the stars?
- Scene 5: Inspiration Astronomy is not just about stars and planets; it is about understanding our place in the cosmos. ○ Scene 6: Explore the Universe. Discover the Unknown.

• Example of Few-Shot Prompting:
- Narration Style Deep, calm, space-documentary voice similar to sci-fi films like Interstellar.
- Visual Style Star fields, nebulae, planets, astronauts, glowing galaxies, slow cinematic camera movement. ○ Transitions Smooth fade-outs, cosmic light effects, slow zoom-ins

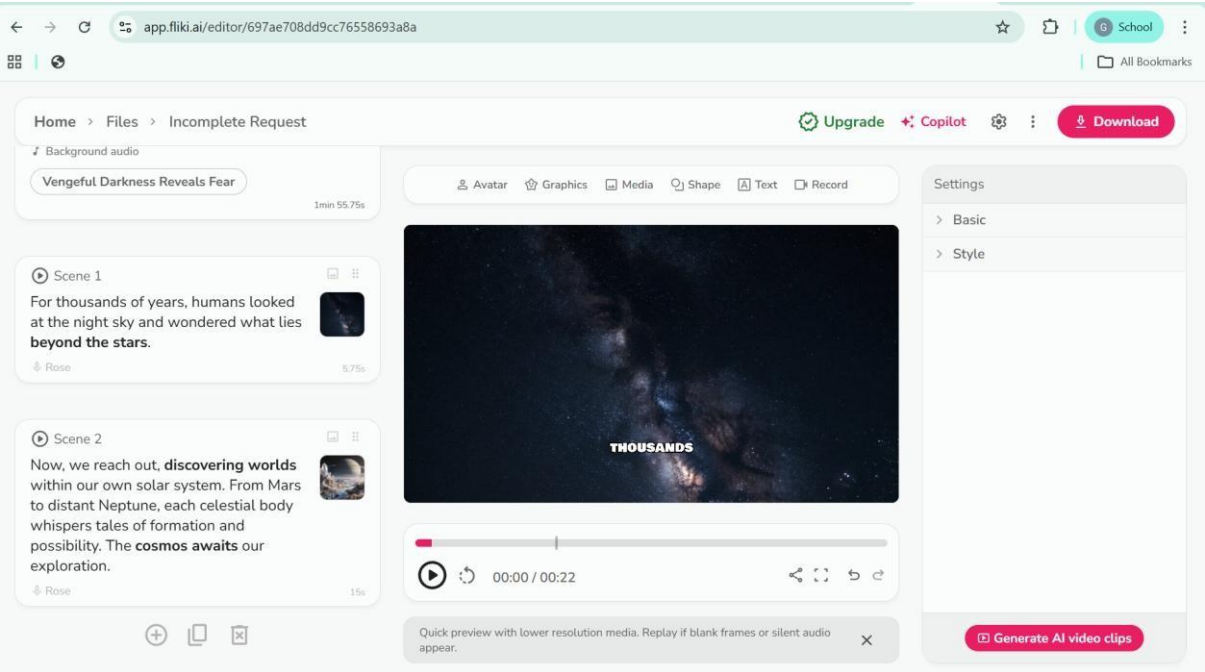
- Video generation using the Chain of thoughts provided in the above example as prompt in fliki.ai







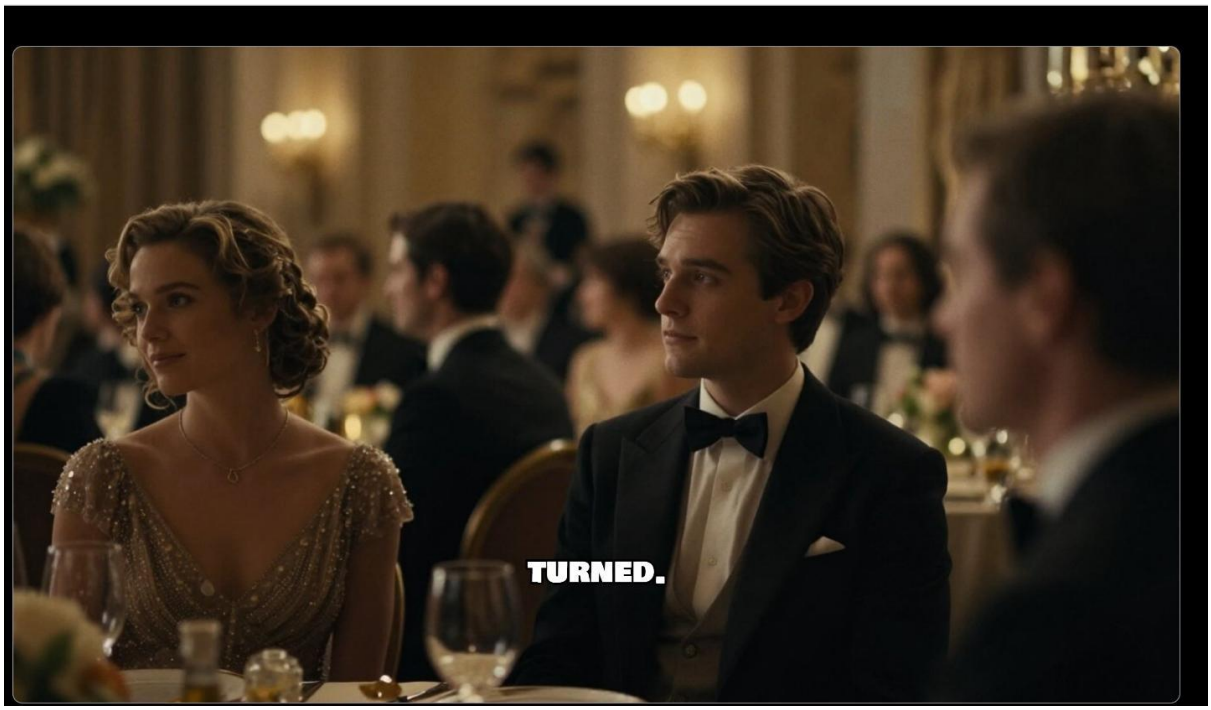
Gagana.K
1RVU23CSE162

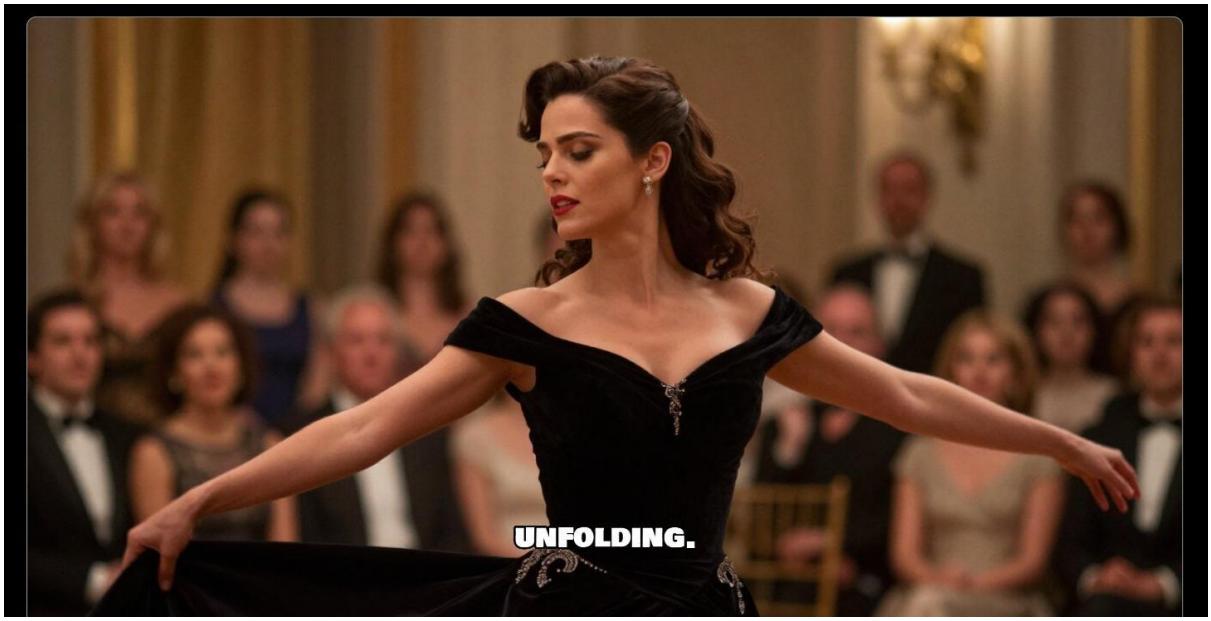


- Prompt: Generate a video of a woman who enters a ball in a beautiful long dress, she draws all the attention with her appearance and shows exceptional dance skills, aligning her soul, remarking that beauty lies within.

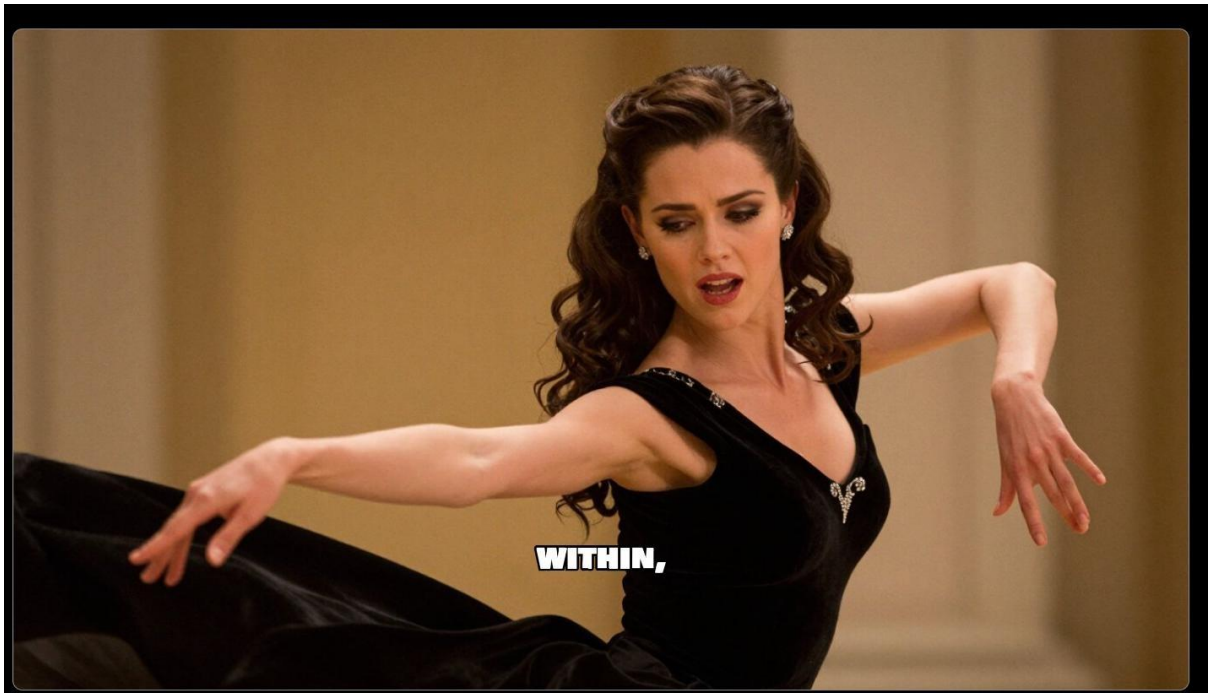


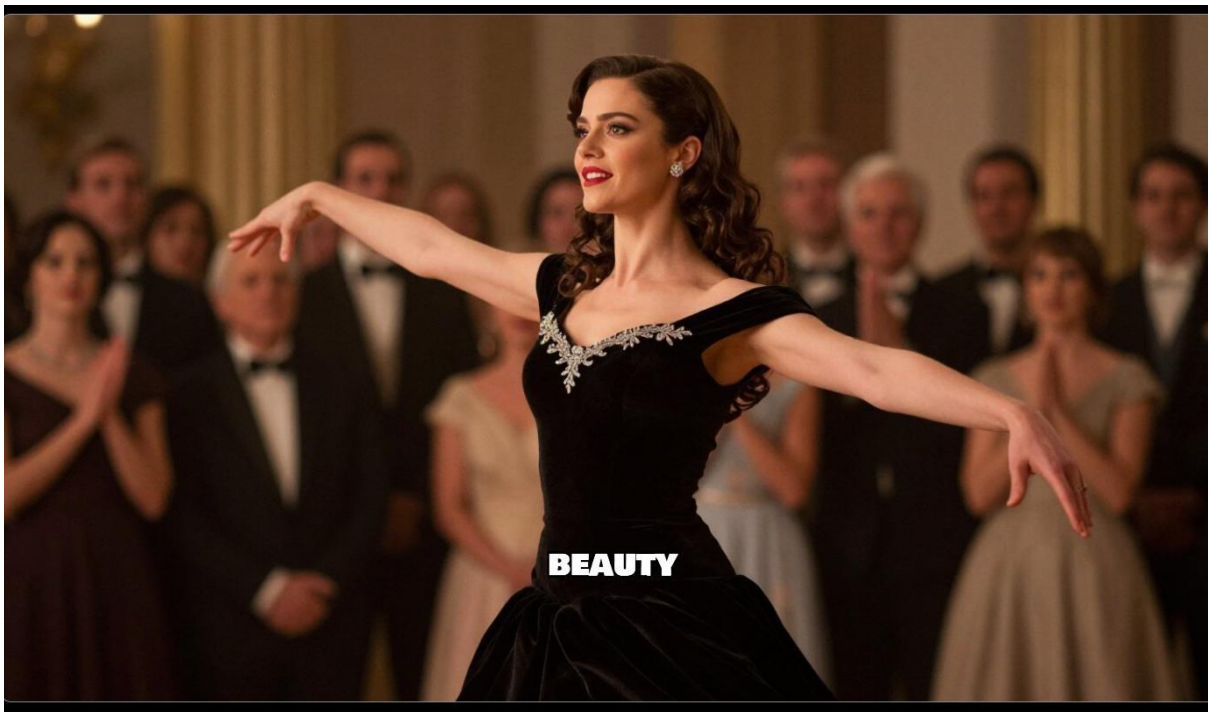
Gagana.K
1RVU23CSE162





Gagana.K
1RVU23CSE162





Gagana.K
1RVU23CSE162